STATISTICAL BUSINESS ANALYSIS

Week 7: Introduction to Statistics for Data Science

Dataset: 500-row customer churn data supplied by the user.

1. Descriptive Statistics

Metric	Mean	Median	Std Dev	Mode
Tenure	36.53	37.00	20.67	3
MonthlyCharges	113.64	115.00	51.80	115
TotalCharges	4237.88	4182.50	2260.62	4023
SeniorCitizen	0.50	0.00	0.50	0
Churn	0.11	0.00	0.31	0

2. Distribution Analysis

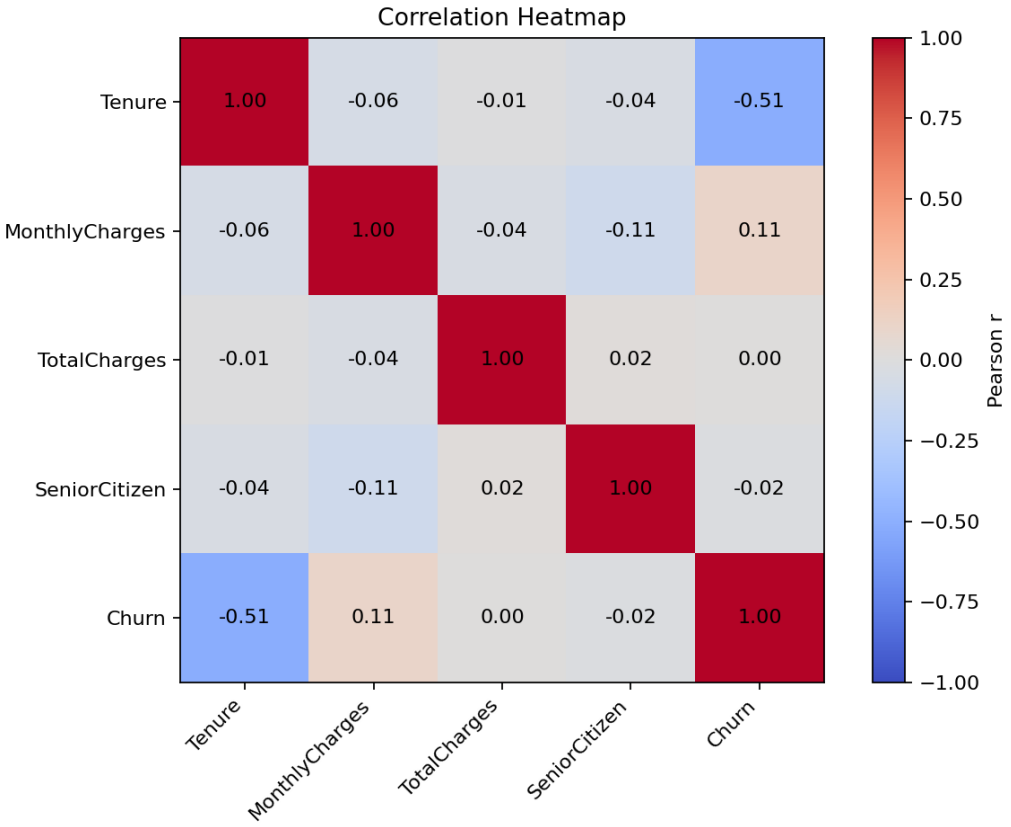
Tenure: Shapiro-Wilk p = 5.10261e-12. Not normal.

MonthlyCharges: Shapiro-Wilk p = 1.14923e-11. Not normal.

TotalCharges: Shapiro-Wilk p = 9.38297e-12. Not normal.

3. Correlation Analysis

Tenure vs Churn: r=-0.509; MonthlyCharges vs Churn: r=0.107; Tenure vs MonthlyCharges: r=-0.060.



4. Hypothesis Testing

Test	Comparison	Statistic	p-value	Result
Welch t-test	MonthlyCharges by Churn	-2.646	0.0100792	Significant
Welch t-test	Tenure by Churn	34.535	8.98233e-130	Significant
One-way ANOVA	MonthlyCharges by Contract	0.031	0.969212	Not significant

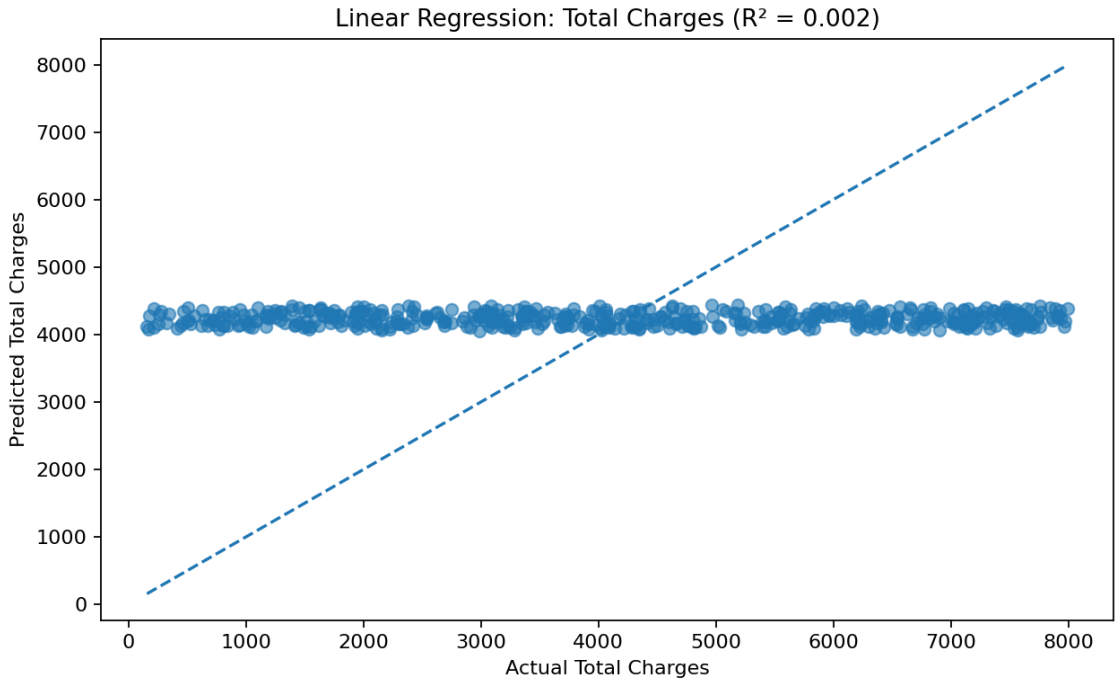
At alpha=0.05, monthly charges and tenure differ significantly between churn groups, while the ANOVA does not detect a significant difference in monthly charges across contract types.

5. Confidence Intervals

Metric	Mean	95% CI	Margin of Error
Tenure	36.53	[34.72, 38.35]	1.82
MonthlyCharges	113.64	[109.08, 118.19]	4.55
TotalCharges	4237.88	[4039.25, 4436.51]	198.63

6. Regression Analysis

TotalCharges was modeled from Tenure and MonthlyCharges. R^2=0.002; overall p-value=0.630398. The model explains very little variation in TotalCharges in this supplied dataset.



7. Business Insights and Limitation

Churn rate is 10.6%. Non-churned customers average 40.15 months of tenure versus 6.00 months for churned customers. Average monthly charges are $111.72 versus $129.77. These are statistical associations and do not prove causation.

Important: the uploaded files are customer-churn datasets, not the assignment's sales_data.csv. They contain no marketing-spend or sales-revenue fields, so that specific analysis is not fabricated here.